

Memory State: Inference-Native Retrieval for Long-Term Agent Memory

Gordon Xiong

Seetie Inc.

Code and results: https://github.com/Seetie-AI/memory_state

Abstract

Agent memory systems typically pair a reasoning model with a separate, smaller embedding model that indexes what the agent should remember. We ask whether the reasoning model can supply its own retrieval keys. Memory State prompts a generative LLM to compress a memory page into a short key, then stores the hidden state at the final key token, taken from a late-layer band near the output, as the retrieval vector. No retriever is trained, distilled, or adapted.

On LongMemEval-S at round granularity, a compact two-view configuration using Qwen3.5-9B-MLX-4bit reaches strict Recall@5 of 0.777 using hidden states alone, at 8 KB per page, against 0.755 for a Qwen3-Embedding-8B-4bit-DWQ baseline on the same 94 scored questions. The margin is two questions on this subset, so we report it as parity-level evidence and make no significance claim.

To test whether prompt semantics carry across benchmarks rather than fitting one of them, we carry the three prompt identities selected on LongMemEval over to PrefEval implicit-persona, a preference and persona memory task, and evaluate them over all 1,000 examples. Per-view layer and geometry settings were assigned on the PrefEval side, so this is prompt-identity transfer rather than a frozen configuration. As pure dense hidden-state retrieval with no lexical or embedding signal it reaches Recall@5 of 0.299 against 0.281 for the same 8B embedding baseline, while ranking last among the four K3 candidates evaluated there. A model-size check under matched conditions gives 0.596 for Qwen3.5-2B-MLX-4bit and 0.691 for Qwen3.5-9B-MLX-4bit, so in this matched comparison the memory representation improves with the reasoning model rather than being capped by a smaller retriever.

1 Introduction

File-based agent memory usually begins with plain text and keyword search. It is cheap, transparent, and easy to inspect. It also has a blind spot: the agent must know what to search for. A memory that matters to the current situation may share none of the words in the current query. This is the unknown unknown problem.

Semantic retrieval addresses it, but it adds a second model. That retriever is usually smaller than the reasoning model, and it was trained for a retrieval objective that may not match the agent-memory domain it is deployed in. If it misses, the more capable model never sees the memory.

This work tests a third arrangement. The model that reads and reasons over a memory also produces the vector that finds it later. If that works, the memory index shares an inference path, a tokenizer, and a semantic space with downstream reasoning, and it needs no separately trained component.

The question came from an anomaly. While sweeping prompt suffixes on PrefEval, a generic prompt asking the model for a single emoji reached Recall@5 of about 0.285, next to 0.281 for an 8B embedding model on the same 1,000 examples. That prompt was not the best in the sweep and it is not part of any system reported here. We record it as the observation that motivated the work: a reasoning model appears to carry a usable memory representation in places we do not normally treat as an index.

We report three things. First, hidden-state keys reach embedding-baseline scale on evidence retrieval. Second, a prompt combination selected on one memory benchmark carries over to a different memory task. Third, the representation improves with model size under matched conditions.

2 Related Work

PromptEOL [1] showed that a one-word prompt can turn the hidden state of the final prompt token into a sentence embedding without fine-tuning.

MetaEOL [2] extended this with multiple meta-task prompt views over the same text, and reported that the final layer is not always the best choice, with a proportional layer-selection strategy giving improvements.

PromptReps [3] brought prompted LLM representations to training-free full-corpus document retrieval, combining the last layer's last-token hidden state with next-token logits into a dense and sparse hybrid, and reported stronger retrieval with larger LLMs.

All-but-the-Top [4] established that removing the common mean and the top principal components from a set of embeddings exposes semantic differences that a shared dominant direction otherwise masks.

Memory State does not claim the mechanism. Prompted final-token hidden states, multiple prompt views, layer selection below the output layer, and dominant-direction removal all come from this line of work. Our contribution is the application: treating these representations as persistent retrieval keys for long-term agent memory, designing memory-specific prompt suffixes, and evaluating the result on two memory benchmarks.

We evaluate on **LongMemEval** [5] and **PrefEval** [6].

3 Method

For each memory page and each query, the model is run over the text plus a retrieval suffix that asks it to compress the text into a short key. The retrieval vector is the hidden state at the final token of that suffix.

The pattern is:

1. choose prompt views that match the memory domain;
2. extract the suffix-final hidden state from a late-layer band near the output;
3. apply dominant-direction removal to strip shared prompt and corpus geometry;
4. combine views by concatenation or component-normalized vector averaging;
5. optionally add a lexical score as a shortlist rerank signal.

Queries are encoded with the same recipe as memory pages. We tested the asymmetric alternative, encoding memory and query with different prompts, and it did not help: the best asymmetric configuration reached Recall@5 of 0.755 against 0.766 for symmetric encoding of the same prompt.

The reported configurations use hidden states as the dense representation. Next-token logits were also collected, and a PromptReps-style sparse path was reproduced, but it is not part of the reported systems.

3.1 Prompt views

Suffixes are written in Chinese and ask the model to emit a single token. One of the PrefEval views, translated, reads:

Use one emoji to mark the emotion of the conversation above. The emoji is: "

The trailing quotation mark is part of the suffix. It opens the model's answer, so the next decoded token is the key itself. The other two views ask for one keyword the conversation should recall and one token for the association it creates. The originals and the full configuration are in the repository.

3.2 Geometry

Dominant-direction removal is applied symmetrically to query and candidate vectors with rank 15. Across 17 symmetric prompt variants, symmetric removal won Recall@5 for 13, query-only removal for 3, and plain centered cosine for 1.

4 Experimental Setup

Hidden-state extraction uses Qwen3.5-9B-MLX-4bit, with Qwen3.5-2B-MLX-4bit for the model-size check. Embedding baselines are Qwen3-Embedding-8B-4bit-DWQ and Qwen3-Embedding-0.6B-4bit-DWQ, from the Qwen3 Embedding series [7]. All models are local 4-bit conversions. Embedding scores are never concatenated into the hidden vectors; they appear only as baselines or as score-level signals.

LongMemEval-S is evaluated at round granularity on a 100-instance subset, of which 94 questions score after the official abstention filter. Recall@5 is strict: a question scores 1 only if every gold turn appears in the top 5. One question moves the metric by about 1.1 percentage points. PrefEval implicit-persona is evaluated over all 1,000 examples.

Benchmark data is not redistributed with the repository. Sources and download instructions are documented there.

5 Results

5.1 LongMemEval

Table 1: Retrieval results on LongMemEval-S.

Method	R@5	NDCG@5	MRR
BM25	0.585	0.547	0.559
Qwen3-Embedding-8B-4bit-DWQ	0.755	0.789	0.826
Memory State, compact K2, hidden only, 8 KB/page	0.777	0.788	0.820
Memory State, K3 concat plus BM25 top 20, 24 KB/page	0.777	0.822	0.851

Metric guide, for this and every table below. R@k (Recall@k) measures how much relevant memory evidence appears within the first k results; LongMemEval uses a strict version requiring every gold turn in the top five. NDCG@5 rewards relevant evidence appearing nearer the top. MRR measures how early the first relevant result appears. Higher is better throughout.

The compact two-view configuration matches the three-view hybrid on strict recall while using no lexical signal and one third of the storage. Its 0.022 margin over the 8B embedding baseline equals two questions in this sample. At that resolution we make no significance claim. This is parity-level evidence at embedding-baseline scale without training an embedding model, not a decisive held-out win.

5.2 Cross-task transfer to PrefEval

The three prompt identities were selected on LongMemEval. They were then evaluated on PrefEval as pure dense hidden-state retrieval over all 1,000 examples. Per-view layer and dominant-direction settings were assigned when the PrefEval K3 candidates were constructed, so what transfers is the prompt combination, not a frozen extraction configuration.

Table 2: Cross-task retrieval results on PrefEval implicit-persona.

Method	R@1	R@3	R@5	NDCG@5	MRR
BM25	0.035	0.074	0.094	0.066	0.067
Qwen3-Embedding-8B-4bit-DWQ	0.093	0.216	0.281	0.191	0.200
Memory State, LongMem prompt transfer, hidden only	0.093	0.212	0.299	0.197	0.188

The transferred combination exceeds the 8B embedding baseline on strict Recall@5 using no lexical signal and no embedding score. It is slightly lower on R@3 and MRR. The LongMemEval shortlist and fusion weights were not carried over.

Two things make this the most informative PrefEval row we have. It was not selected to win here: of the four K3 candidates evaluated on PrefEval it placed last, below combinations whose prompts were chosen on PrefEval itself. And its prompt identities were fixed by a different benchmark. So it is the weakest candidate in its own sweep and still clears the 8B embedding baseline on strict recall. This supports the claim that a prompt combination can cross memory domains, not that every low-level parameter is universal.

A separate configuration tuned directly on PrefEval, fusing the hidden-state score with an embedding score and BM25, reaches Recall@5 of 0.332. Because its prompts and fusion weights were selected on the same data it is reported on, we treat it as a domain-tuned exploratory result and not as the headline.

5.3 Model size

Both models were run with the same prompt suffix, the same centered transform, the same 4-bit precision, and the same 100 instances.

Table 3: Matched model-size comparison.

Model	Layer	R@5	NDCG@5
Qwen3.5-2B-MLX-4bit	22	0.596	0.547
Qwen3.5-9B-MLX-4bit	30	0.691	0.689

The larger model gains 9.5 percentage points on Recall@5. On a smaller 30-instance tuning slice both models scored 0.833, so the gap appears only at the larger sample. Each model uses its own best late layer rather than a fixed absolute index. The direction is consistent with earlier retrieval results reported for prompted representations at larger model scale [3].

6 Analysis and Negative Results

There is no universal best layer. Across the prompt sweep, layers 29, 30, and 31 each won for 6, 6, and 5 variants respectively. The useful region is a late-layer band whose exact position depends on prompt semantics, not a single layer.

Prompt wording matters more than expected. Changing one word in an otherwise identical suffix, from a neutral term for the other party to the product-native term for the user, moved Recall@5 by 0.043. The model appears better aligned to product vocabulary than to neutral phrasing.

Prompt language is not neutral. In the sweep, the best cell for an English variant of one suffix reached Recall@5 of 0.713 while the best cell for its Chinese original reached 0.755, on English memory content. The two best cells do not share a layer and transform, so this is a sweep observation rather than a single-variable comparison. Cross-language prompting worked here; we make no general multilingual claim.

Not every retrieval-shaped prompt works. An answer-strategy prompt fell to Recall@5 of 0.574.

Lexical fusion helps only as a small vote inside a shortlist. Fusing BM25 inside the vector top 20 or top 50 beat full-corpus fusion for ranking, and weighting BM25 heavily collapsed performance across configurations, because turn-level lexical matching is noisy.

Late interaction did not justify its cost. Max-sim style scoring over multiple stored vectors did not beat the simpler concatenation and averaging candidates.

Agreement filtering is a diagnostic, not a default. Gold candidates were much more likely than non-gold candidates to appear in multiple prompt rankings, but sorting by agreement never beat sorting by score.

A stronger reranker buys ranking, not recall. Blending an 8B embedding score into the hybrid raised NDCG@5 to 0.834 and MRR to 0.888 without improving Recall@5, while reintroducing the separate model dependency the method exists to avoid. We keep it as a ceiling, not a default.

Small subsets misled us. An early prompt that led a 40-instance slice fell to mid-pack on the full 100 instances. Oracle union diagnostics, which use gold labels to find complementary prompt pairs, were likewise treated as hypothesis generators rather than estimates.

7 Deployment Considerations

The vector comes from the final token of a prompted key-generation pass. In an online setting the reasoning model has already processed the conversation and its context is in the KV cache, so capturing a memory key means decoding one additional token per prompt view and keeping its hidden state. Memory capture then happens inside the inference path rather than as a second encoding job by another model. We did not measure end-to-end latency or cost, so this is a deployment argument, not a measured result.

Storage is straightforward. Qwen3.5-9B has a hidden size of 4096, so one view in bf16 costs 8 KB per page and three views cost 24 KB. Around 20,000 memory pages need roughly 500 MB before index overhead. The compact configuration in Section 5.1 reaches the same strict recall as the three-view hybrid at 8 KB.

8 Limitations

The LongMemEval numbers rest on 94 scored questions, where one question moves the metric by about 1.1 percentage points. Differences of one or two questions between the reported systems and the embedding baselines should be read as parity, not as ordering.

Prompt and fusion choices were selected on explored benchmark subsets. The LongMemEval results are therefore exploratory rather than a clean held-out evaluation. The PrefEval result in Section 5.2 is the closest we come to an independent estimate, because its prompt identities were fixed by a different benchmark and it was not the configuration selected to perform best there. Its per-view layer and geometry settings were still assigned on the PrefEval side, so it is not a held-out evaluation either.

The geometry is tied to Qwen3.5-9B-MLX-4bit and may change under another model, another quantization, another language, or another prompt template. Both models tested come from one family, so we cannot say whether the effect holds across model families.

The work supports feasibility at embedding-baseline scale. It does not establish a general win over embedding models.

9 Reproducibility

Code, experiment logs, the earlier internal results overview, and the result files for all reported numbers are public at https://github.com/Seetie-AI/memory_state.

Benchmark datasets and model weights are not redistributed. The repository documents the original sources for LongMemEval and PrefEval, the download commands, and the model identifiers, so that users obtain them under the original terms.

10 Conclusion

A generative model, in the course of reading a page, leaves behind a representation that can be used to find that page again. On two different memory benchmarks this representation performed at the scale of embedding models built for the job, without training a retriever, and it improved with model size under matched conditions.

The claim is not that hidden states beat embedding models. It is that an agent's memory index may not require a second model at all, and that the memory an agent needs may already be present in the act of understanding, waiting to be read out rather than rebuilt.

References

- [1] Jiang et al. Scaling Sentence Embeddings with Large Language Models. *Findings of EMNLP*, 2024. <https://aclanthology.org/2024.findings-emnlp.181/>
- [2] Lei et al. Meta-Task Prompting Elicits Embeddings from Large Language Models. *ACL*, 2024. <https://aclanthology.org/2024.acl-long.546/>
- [3] Zhuang et al. PromptReps: Prompting Large Language Models to Generate Dense and Sparse Representations for Zero-Shot Document Retrieval. *EMNLP*, 2024. <https://aclanthology.org/2024.emnlp-main.250/>
- [4] Mu and Viswanath. All-but-the-Top: Simple and Effective Postprocessing for Word Representations. *ICLR*, 2018. <https://openreview.net/forum?id=HkuGJ3kCb>
- [5] Wu et al. LongMemEval: Benchmarking Chat Assistants on Long-Term Interactive Memory. 2024. <https://arxiv.org/abs/2410.10813>
- [6] Zhao, Hong, Liu, Hazarika, and Lin. Do LLMs Recognize Your Preferences? Evaluating Personalized Preference Following in LLMs. 2025. <https://arxiv.org/abs/2502.09597>
- [7] Zhang et al. Qwen3 Embedding: Advancing Text Embedding and Reranking Through Foundation Models. 2025. <https://arxiv.org/abs/2506.05176>